

NOAA Hurricane Predictions

EGN5442: Term Project Report

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Abstract—This report details the cleaning, preparation, and modeling of the NOAA storm dataset to better understand and predict the behavior of hurricanes, focusing on baseline classification models and advanced time-series predictions. We transformed raw data into a structured format to support machine learning models, performed exploratory data analysis through visualizations, and utilized RandomForest for storm classification. Ultimately, we implemented a SARIMAX model to predict hurricane wind speeds over time. Our results show high accuracy in classifying hurricanes, but reveal opportunities for improvement with lesser storm categories and future time series implementations.

I. INTRODUCTION

Hurricanes and tropical cyclones are among the most damaging natural hazards affecting coastal communities. Accurate historical analysis and forecasting of hurricane intensity and track are critical for preparedness and mitigation. This project uses historical NOAA storm records to explore patterns in hurricane behavior, build baseline classification models to categorize storm strength, and implement time-series forecasting to predict wind speeds during selected storms. The work combines data cleaning, exploratory visualization, classification with Random Forests, and time-series modeling using SARIMAX.

The remainder of the report is organized as follows: Section II describes the dataset and preprocessing steps; Section III presents exploratory data analysis and visualizations; Section IV presents baseline classification models; Section V describes the SARIMAX time-series approach and predictions for selected storms; Section VI discusses results and limitations; and Section VII concludes with recommendations for future work.

II. DATA CLEANING AND PREPARATION

We used the NOAA storm dataset and focused on cleaning and preparing the data to ensure it was suitable for machine learning models. First, we used `.info()` and `.head()` functions to get an understanding of the dataset structure. Our dataset contains 19,066 entries with 14 columns. We found significant amounts of missing data in the ‘category’ and ‘diameter’ columns of the dataset because any status that was not a hurricane would not have a category label; to address this issue, we included the category within the status variable making “hurricane” into 5 separate status variables dependent upon category.

Our next step in data processing was to convert date/time columns to datetime objects and remove redundant date columns. Additionally, we handled excess categorical variables in the status column such as “tropical wave” by condensing multiple categories into 10 distinct groups, considering outlying status variables as ‘disturbance’. We also trained and split data with specifications to prepare for the RandomForest model training using the `train_test_split` function in the `sklearn` Python package; this function trains the initial model and checks the model generalization and performance. Overall, we made an effort to transform our raw data into a more structured format based on model characteristics.

III. DATA EXPLORATION AND VISUALIZATIONS

After cleaning the data, we utilized visualizations to understand the feature distributions and explore the relationship between them; this gave us a better understanding of the data within the dataset to prepare for writing models. First, we calculated a correlation matrix and created a heatmap to visualize the correlations between each of the variables within the dataset. Initially, the goal of the correlation matrix was to look for correlations with the latitude and longitude variables in particular, but no strong associations were found.

The associations that were found within the data were expected based on current knowledge of weather patterns. For example, a 93% negative correlation was found between pressure and wind, indicating that as pressure decreases wind speed increases; this association was demonstrated using a scatter plot. Additionally, a 77% and 64% negative correlation was found between pressure and hurricane force diameter and pressure and tropical storm force diameter, respectively. To visualize this relationship, a scatter plot was created with both tropical storm force diameter and hurricane force diameter on the y-axis; combining these variables into one graph allowed for a comparison across all status variables.

After examining this scatter plot further, we noticed that the manually combined disturbance variable had a large distribution of points across wind speed and force diameter. To better understand the range of wind speeds by storm status, a boxplot was created. This plot clearly shows that there was a wide distribution of wind speed points within the disturbance status; average wind speed ranged from 15 kt to 105 kt with a median of about 58 kt. This calls into question our decision to combine outlying status variables into one status variable;

this decision could potentially affect model performance, but it was useful for data cleaning.

Next, we utilized `folium` to create maps to demonstrate the geographical spread of the storms within the dataset. Using latitude and longitude as well as storm wind speed data, we created a geographical distribution of hurricanes included within the dataset. Latitude and longitude data was used to map the geographical distribution of hurricanes. Peak positions were defined based on maximum wind speeds and lowest pressures recorded for each storm. The map revealed concentrated hurricane activity along the Gulf of Mexico and the Atlantic coastline with higher wind speeds further south which aligns with currently established hurricane pathways and regions of vulnerability.

From here, we were interested in filtering the data to display only Florida-based storms, so we filtered the dataset to include entries where the latitude and longitude values fell within geographic bounding boxes specific to Florida’s coordinates, for instance latitude range 24.396308 (southernmost point) to 31.000888 (northernmost point) and longitude range -87.634896 (westernmost boundary) to -80.031362 (easternmost boundary). Further refinements included isolating records marked as hurricane in the dataset’s status column. According to the map, the majority of Florida-bound storms originated in the Gulf of Mexico and the Atlantic Ocean, following well-established storm paths. Storm intensity diminished significantly upon landfall due to the lack of warm ocean waters that typically fuel hurricanes. This phenomenon was evident from declining wind speeds and pressure changes post-landfall, visualized in scatter plots and intensity maps. To support this map, we generated a scatter plot showing the latitude and longitude coordinates of each named hurricane that made landfall in Florida.

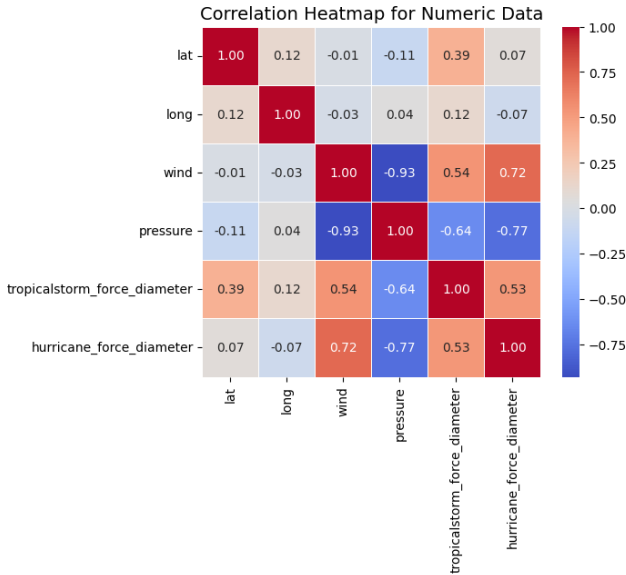


Fig. 2: Florida-focused storm tracks and landfall points (extracted).

IV. BASE MODEL DEVELOPMENT

To begin work on model development, we started with a base model to validate our data pipeline and establish a baseline. For our base model, we chose RandomForest because of its robustness and versatility; RandomForest is able to handle non-linear relationships while still offering a more simplistic prediction method. We trained and ran a model without hyperparameter tuning with 20% of the data used for the test set and randomization set at 44. We then calculated precision, recall, f1-score, and support as well as a confusion matrix.

Our results showed that hurricanes were accurately predicted with this base model with precision and recall values from 0.99 to 1.00. However, other types of storms (e.g. subtropical storm, subtropical depression) were less accurate within this model most likely due to a lack of data points. For example, a subtropical depression only had 151 points within the dataset.

V. ADVANCED MODEL DEVELOPMENT

In applying more advanced models to the data, we chose to complete a time series analysis using SARIMAX (Seasonal Autoregressive Integrated Moving-Average with eXogenous regressors). SARIMAX utilizes three basic features of ARIMA (Autoregressive Integrated Moving Average): autoregression (AR), moving average (MA), and integration (I).

AR predicts current values based on current values, it utilizes an order denoted with p . AR computes a linear regression on the last p values of the time series. For this model p was set at 1. Moving Average, similarly, deals with past errors in predictions. It represents an order with a value of q . MA similarly computes a linear regression on the last q errors. Our model set q at 1. Integration corresponds to the number of differentiations in the data required to make the time series “stationary”. We chose this value, represented by d to be 1.

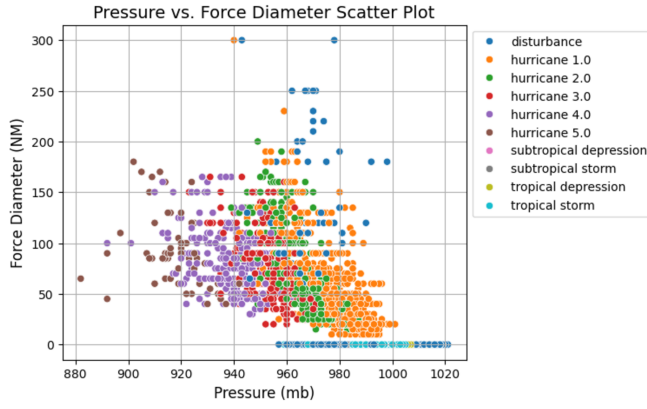
$$\text{Autoregression Equation: } X_t = \varepsilon_t + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p}$$

$$\text{Moving Average Equation: } X_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q}$$

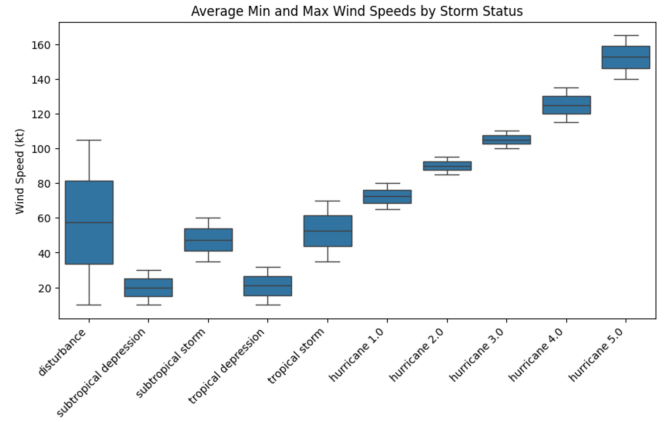
To include the pattern of seasonality found in all hurricanes, the SARIMAX model adds a ‘seasonal_order’ component to the model. This adds a second set of $\{p, d, q\}$ coefficients plus an $\{s\}$ coefficient corresponding to the seasonal period of the data. These were set at $\{2, 1, 0, 6\}$ respectively. In our dataset there were a total of 258 unique storms. Among these were some highly documented ones (Michael, Irma, Katrina) while others only contained a few data points. For the purposes of modeling, only storms which contained at least 10 data points were considered for modeling. After filtering, SARIMAX models were run for each hurricane in that data, this consisted of 187 hurricanes.

A. Hurricane Irma and Michael Predictions

To test the training model, test predictions were run to estimate the wind speeds during Hurricane Irma in 2017 and



(a) Geographical distribution of storms (extracted).



(b) Correlation heatmap (extracted).

Fig. 1: Exploratory visualizations: storm map and correlation heatmap.

Hurricane Michael in 2018. Predictions were computed using the filter method of the statsmodels Python package. This uses Kalman filtering (linear quadratic estimation) to evaluate the log-likelihood for each datetime in the time-series.

VI. RESULTS

This section summarizes the quantitative and qualitative results from the classification and time-series experiments.

extbfClassification (Random Forest). The Random Forest baseline produced very high accuracy for clearly-defined hurricane classes: precision and recall for the hurricane category were approximately 0.99–1.00 (see the classification diagnostic in Figure 5). The exploratory boxplots in Section III show the distribution of wind speeds by status and help explain why the hurricane class is well-separated from other categories. However, performance on lesser or more ambiguous categories (e.g., subtropical storms, depressions) was noticeably worse: subtropical storm recall was low (about 35%), indicating many subtropical events were misclassified as other categories. This is likely due to class imbalance and the fact that some categories are defined by characteristics not fully present in the dataset.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19066 entries, 0 to 19065
Data columns (total 14 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Unnamed: 0          19066 non-null  int64
1   name                19066 non-null  object
2   year                19066 non-null  int64
3   month              19066 non-null  int64
4   day                19066 non-null  int64
5   hour               19066 non-null  int64
6   lat                19066 non-null  float64
7   long               19066 non-null  float64
8   status              19066 non-null  object
9   category            4684 non-null   float64
10  wind                19066 non-null  int64
11  pressure            19066 non-null  int64
12  tropicalstorm_force_diameter  9554 non-null   float64
13  hurricane_force_diameter  9554 non-null   float64
dtypes: float64(5), int64(7), object(2)
memory usage: 2.0+ MB
```

Fig. 5: Classification diagnostics and confusion matrix (extracted).

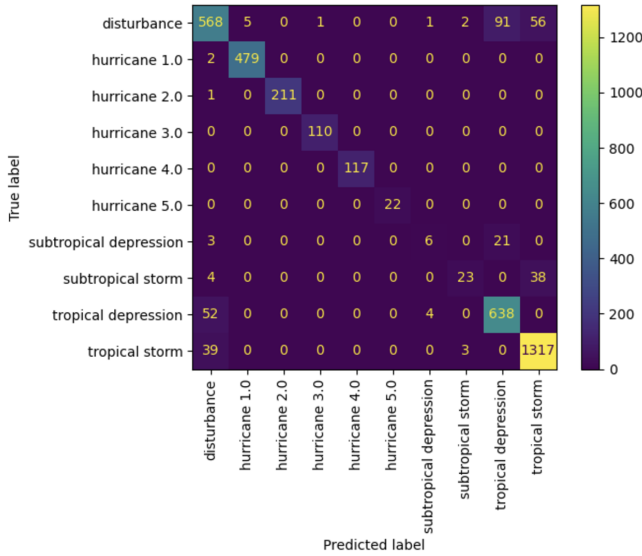
extbfTime-series (SARIMAX). The SARIMAX models trained per-hurricane produced useful short-term forecasts of wind speed. Figure 3a (left panel of the time-series predictions) shows the model prediction for Hurricane Irma: the model captures the overall trend and the primary peak, though short-duration peak magnitudes may be slightly under- or over-estimated. Figure 3b (right panel) shows similar behavior for Hurricane Michael: the predicted mean follows the observed series closely and gives an actionable short-term estimate of wind intensity.

Figure 4a provides a broader wind-speed timeline used for training and demonstrates how the model fits vary across time, while Figure 4b displays residuals and diagnostic plots. The residuals indicate a near-zero mean bias but reveal remaining autocorrelation at short lags in some storms, suggesting the AR/MA orders or exogenous inputs could be further tuned. Overall, SARIMAX gives reasonable forecasts for storms with sufficient data (we used storms with ≥ 10 records), but

	precision	recall	f1-score	support
disturbance	0.85	0.78	0.82	724
hurricane 1.0	0.99	1.00	0.99	481
hurricane 2.0	1.00	1.00	1.00	212
hurricane 3.0	0.99	1.00	1.00	110
hurricane 4.0	1.00	1.00	1.00	117
hurricane 5.0	1.00	1.00	1.00	22
subtropical depression	0.55	0.20	0.29	30
subtropical storm	0.82	0.35	0.49	65
tropical depression	0.85	0.92	0.88	694
tropical storm	0.93	0.97	0.95	1359
accuracy			0.92	3814
macro avg	0.90	0.82	0.84	3814
weighted avg	0.91	0.92	0.91	3814

(a) SARIMAX prediction: Hurricane Irma (extracted).

Fig. 3: Time-series predictions for selected hurricanes.



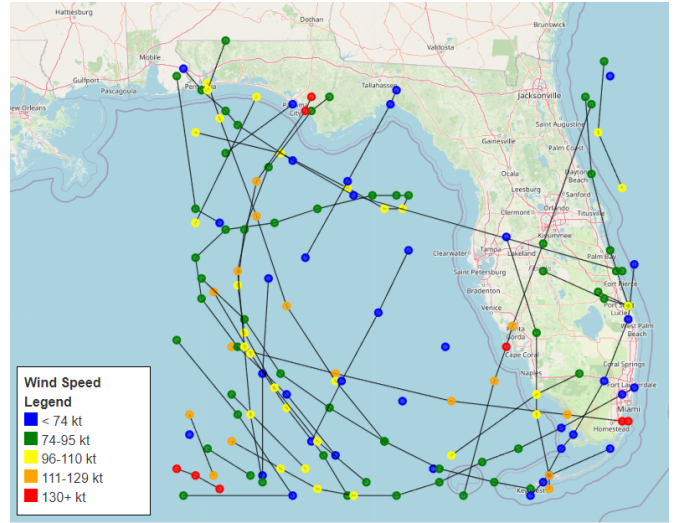
(a) Wind speed timeline (extracted).

Fig. 4: Additional SARIMAX / time-series diagnostics.

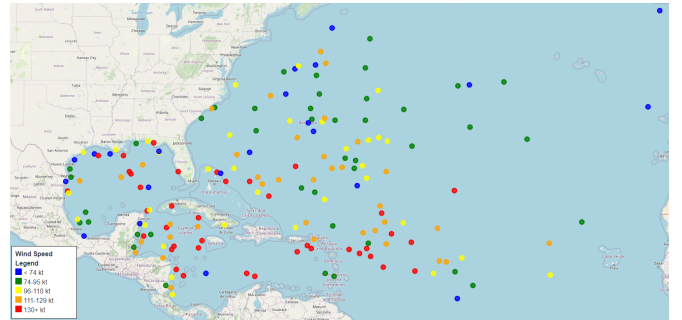
performance degrades for sparsely-sampled storms.

VII. DISCUSSION

For the classification problem using a Random Forests Classifier, the model was very accurate at predicting the strength of hurricanes based on wind speed (precision and recall at 99% or 100%). This comes at no surprise given there is a standardized scale for hurricane classification (Saffir-Simpson Hurricane Wind Scale). However, for lesser storm classification, this model did not result in the greatest accuracy. Subtropical storms had a precision of 82% which is not horrible for a non-tuned model, but the recall was as low as 35%, over 70% of the time incorrectly classifying a subtropical



(b) SARIMAX prediction: Hurricane Michael (extracted).



(b) Model residuals and diagnostics (extracted).

storm. This could very well be because of the low number of these types of storms in the dataset, but also could be caused by the varied classification of subtropical storms as a whole. These storms, unlike their hurricane counterparts, are not always purely classified based on wind speeds and take into account prior and current barometric pressure, diameter of these readings based on the computed center of the cycle, and many other factors. These are not all represented in the dataset so this base model is unlikely to predict the categorizations of the storms with the data it is trained with alone. Better data, and a larger data set might improve this model's accuracy, as well as trying a more advanced classification model such as

neural networks or SVM.

Results from the SARIMAX time-series model showcase that this type of model is relatively accurate in the prediction of wind speeds over the course of a hurricane's life. Further accuracy might be obtained by training the model on more hurricanes given only 187 were used in the training of our model. Utilizing a larger dataset such as HURDAT2 or in combination with other data sets, may give our model more accurate predictions. Additional features which might improve the models prediction of wind speeds could include current air and water temperatures around the eye, whether the current location is over land or water, and possibly percentage of overall global warming when compared historically. Further use of the model could be during active hurricanes to predict upcoming wind speeds based on these previous and currently measured data points. Combining this with newer modeling options, such as those using computer vision, or other forms of advanced machine learning classification, might produce more accurate models for predicting the future strength of active hurricanes.

VIII. CONCLUSION

This project demonstrates a reproducible workflow to analyze NOAA storm data, produce interpretable visualizations, train a baseline classifier, and implement per-storm SARIMAX forecasts. Key takeaways:

- A Random Forest baseline reliably identifies hurricane-class events where wind-speed separation is strong, but struggles with under-represented or more nuanced categories.
- SARIMAX provides useful short-term wind-speed forecasts for storms with moderate-to-high data density, capturing primary trends and peaks but leaving room for improvement in peak magnitude prediction and residual autocorrelation.

Future work should focus on enlarging the training data (HURDAT2 and other sources), adding physically-relevant exogenous variables (sea-surface temperature, land/water mask), and exploring ensemble and deep-learning approaches for both classification and forecasting tasks.

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